

ADRIAN MATIAS

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PROFILE

Data Scientist / ML Engineer with 12 years shipping end-to-end machine learning from problem framing to production. Deep roots in core DS — ad-tech ranking & user acquisition (Appodeal, Chartboost), fraud detection, NLP (RavenPack), recommender systems, Bayesian Bandits / RL, and large-scale experimentation on Spark / Databricks — with current work in GenAI: document extraction, scoring engines, and LLM-driven platforms on Azure OpenAI (PwC Advisory via Cavendish). Owns design → deployment → monitoring across client engagements. Approach driven by simplicity, first-principles thinking, and measurable business outcomes.

EXPERIENCE

Senior AI Engineer

Cavendish Professionals → PwC Advisory (long-term contract) | León, Spain – Remote | May 2026 – Present

Backend / analytic-engine engineer on a cross-timezone pod building an LLM-driven Vendor Evaluation Tool (VET) for PwC Advisory: ingest heterogeneous vendor submissions → extract → score → compare. Full-time engagement through Cavendish Professionals. Stack: Python, FastAPI, Pydantic, PostgreSQL, Azure OpenAI, Next.js, GitHub Actions; deployed on internal AKS.

KEY ACHIEVEMENTS

- ▶ Shipped typed FE/BE response contract (Pydantic as source of truth) across a 3-PR arc, with a CI gate that fails on camelCase/snake_case wire drift — closing a recurring class of silent FE/BE integration bugs.
- ▶ Made the backend the single source of truth for derived scores (out of raw LLM output); diagnosed and fixed a weighting defect where headline vendor score ignored framework dimension weights.
- ▶ Delivered document-level provenance (every extracted value traces to source) and a local e2e test harness for reproducible assessment runs outside the deployed stack.
- ▶ Ran a market-scan / vendor-crawl spike end-to-end to validate the generic LLM scoring path and surface the weighting defect.
- ▶ Correctness reviewer on peers' PRs (load-bearing scoring/framework bugs caught pre-merge). Delivery: 16 authored PRs in ~3 weeks (11 merged).

Data Scientist → DS Technical Lead

Appodeal: appodeal.com | León, Spain – Remote | Dec 2024 – Jan 2026

Developed install prediction models for mobile user acquisition: cold-start, transformers, user embeddings, transfer learning, MLOps, production pipelines, parity resolution, experimentation, and monitoring. Progressed toward technical leadership based on ownership and strategic perspective.

KEY ACHIEVEMENTS

- ▶ Unified modeling across targets, advertisers, and platforms via attention/transformer architecture, increasing end KPI.
- ▶ Enabled seamless advertiser onboarding through novel cold-start techniques (custom PyTorch + transfer learning).
- ▶ Resolved parity mismatches and calibration gaps; productionized parity workflows with unit tests and decoupling in Databricks.
- ▶ Transfer-learning user embeddings achieving performance parity with orders-of-magnitude cost savings.

Senior Data Scientist

Quant AI Lab: quant.global | León, Spain – Remote | Jul 2024 – Dec 2024

Designed and delivered end-to-end ML solutions for international clients; reviewed technical scope, aligned with stakeholders, and supervised model productivization.

KEY CONTRIBUTIONS

- ▶ Led rapid turnaround of underperforming client ML system (utility fraud detection & ranking): diagnosed architectural issues, executed a surgical plan, restored accuracy/efficiency, and recovered client satisfaction.
- ▶ Discovered critical issues in data ingestion and feature-engineering pipelines.
- ▶ Led real-time credit card fraud classifier combining scalable ML with deep VAE. Open version: github.com/adrianmatias/ml-sandbox/tree/master/tx_class

Senior Data Scientist

Chartboost: chartboost.com | León, Spain – Remote | Jun 2023 – Jun 2024

Developed ad ranking and reinforcement learning systems for in-app game advertising; advanced DSP performance through large-scale experimentation and behavioral analysis.

KEY ACHIEVEMENTS

- ▶ Evolved and monitored Bayesian Bandit RL system with policy evaluation tools; designed tailored online/offline evaluation frameworks.
- ▶ Built Two-Towers neural ranking algorithm based on recommender systems.
- ▶ Post-mortem analysis of adversarial DSP dynamics and model feedback loops.

Machine Learning Engineer

RavenPack | León, Spain – Remote | Mar 2022 – Jan 2023

Researched, implemented, and maintained text analytics, NLU, and semantic search products on massive financial news data; productionized prototypes with emphasis on MLOps and reproducibility.

KEY ACHIEVEMENTS

- ▶ Few-shot search personalization: simplified architecture, reduced error 30%.
- ▶ Active learning PoC for learn-to-rank: projected 300% annotation efficiency gain.
- ▶ Novel long-text fingerprinting algorithm for duplicate content detection.

EARLIER ROLES

Numbrs Aug 2020 – Oct 2021

Extended Deep Learning / Active Learning for transaction processing; scaled classifier across countries with GDPR-compliant pipelines.

Stratio BD Aug 2019 – Aug 2020

Propensity/churn on Spark; BERT+graph email segmentation; multi-source weather DB (Elasticsearch/Flask).

Digilant Jun 2015 – Apr 2017

Redesigned Consumer Persona ML model; Game Theory attribution prototype; co-directed Master's thesis.

IEBS School Oct 2019 – Oct 2021

Webinars and masterclasses on real-time Big Data, AI, and ML applications.

DOCOMO DIGITAL Apr 2017 – Aug 2019

Led fraud detection (99.8% specificity, <50 ms) and default-payment filtering; mentored juniors.

2007–2015

Civil/tunnel engineering; university teaching: Math, Physics, Chemistry.

FREELANCE

Freelance Data Scientist / ML Engineer | Mar 2022 – Jun 2025

On-demand projects in semantic search, time series forecasting, and portfolio optimization for financial analysis.

SKILLS

Core Technologies

Python, FastAPI, Pydantic, PostgreSQL, PyTorch, Transformers, Azure OpenAI / Azure / AKS, AWS/S3, Databricks, PySpark, Pandas, SQL, Docker, MLflow, Git, GitHub Actions, AI-augmented tools (Ollama, llama.cpp, Hermes Agent)

Engineering Practices

Clean code, typed contracts as source of truth, CI drift gates, OOP & FP, TDD / local e2e harnesses, reproducible pipelines, Agile / XP, production ownership

Machine Learning & GenAI

LLMs, RAG, agentic systems, LangChain / LangGraph-style orchestration, document extraction & provenance, scoring engines, Deep Learning (attention, embeddings, transfer learning), Recommender Systems, NLP, RL / Bayesian Bandits, MLOps (parity, A/B, monitoring), Few-Shot / Active Learning, Ranking

Consulting & Soft Skills

Multi-client consulting delivery, stakeholder communication, PR correctness review, mentoring, business impact focus, simplification / unification

SELECTED COURSES & CERTIFICATIONS

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|---|--|
| - Claude Code: A Highly Agentic Coding Assistant: Coursera (2026) | - Functional Programming in Scala Specialization: Coursera (9.2/10, 2017) |
| - ML Engineering for Production (MLOps) Specialization: Coursera (2021) | - Bitcoin and Cryptocurrency Technologies: Coursera (9.5/10, 2017) |
| - Deep Learning with PyTorch: Siamese Networks (2022) | - Algorithms & Data Structures with Java (I, II): Coursera (8.5/10, 2014) |
| - Deep Learning Specialization: Coursera (2016) | - Analytics Edge: Data Science & Predictive Models: MITx (9.3/10, 2014) |
| - Structuring Machine Learning Projects: Coursera (2018) | - Introduction to Programming & Data Science Using Python (I, II): MITx (8.7/10, 2014) |
| - Apache Spark for Big Data & Scalable ML: Coursera (9.9/10, 2015) | |
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SELECTED PROJECTS

- **not the tao with adrian matias:** youtube.com/@notTheTaoWithAdrianMatias Podcast exploring simplicity, consciousness, reality, machine learning, coding, sovereign AI.
 - **RAG Blog System:** github.com/adrianmatias/ml-sandbox/tree/master/ragblog Retrieval-augmented generation pipeline integrating LLMs, vector embeddings, and web crawling.
 - **The Mirror:** <https://github.com/adrianmatias/the-mirror> Knowledge repository based on llm-wiki for personalized assistant and insight for agentic AI.
 - **Credit Card Transaction Fraud Classifier:** github.com/adrianmatias/ml-sandbox/tree/master/tx_class Real-time fraud detection API (classical + unsupervised signals). Deploy-ready: FastAPI, Docker, RayServe.
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EDUCATION

- **Master's in Geological Engineering:** University of Oviedo (2010–2012)
 - **Degree in Mining & Energy Engineering:** University of León (2012)
 - **Bachelor in Mining Engineering:** University of León (2003–2007)
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References available on request